CS121P Homework Assignments

Homework 01

Deliverable: Homework01.py

References:

- https://runestone.academy/runestone/books/published/thinkcspy/GeneralIntro/toctree.html

Sample Input/Output:

```
$ python3 Homework01.py

TTTTT  M    M  U    U
  T    MM  MM  U    U
  T    M  M M  U    U
  T    M    M  U    U
  T    M    M   UUU
```

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Homework 02

Deliverable: Homework02.py

References:
- https://runestone.academy/runestone/books/published/thinkcspy/SimplePythonData/toctree.html
- https://runestone.academy/runestone/books/published/thinkcspy/Debugging/toctree.html

Sample Input/Output:

```
$ python3 Homework02.py
Please enter the number of kilograms you wish to convert: 1
1.0 kilograms is 2 pounds and 3.2739619495804106 ounces.

$ python3 Homework02.py
Please enter the number of kilograms you wish to convert: 0.45359237
0.45359237 kilograms is 1 pounds and 0.0 ounces.
```

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Homework 03

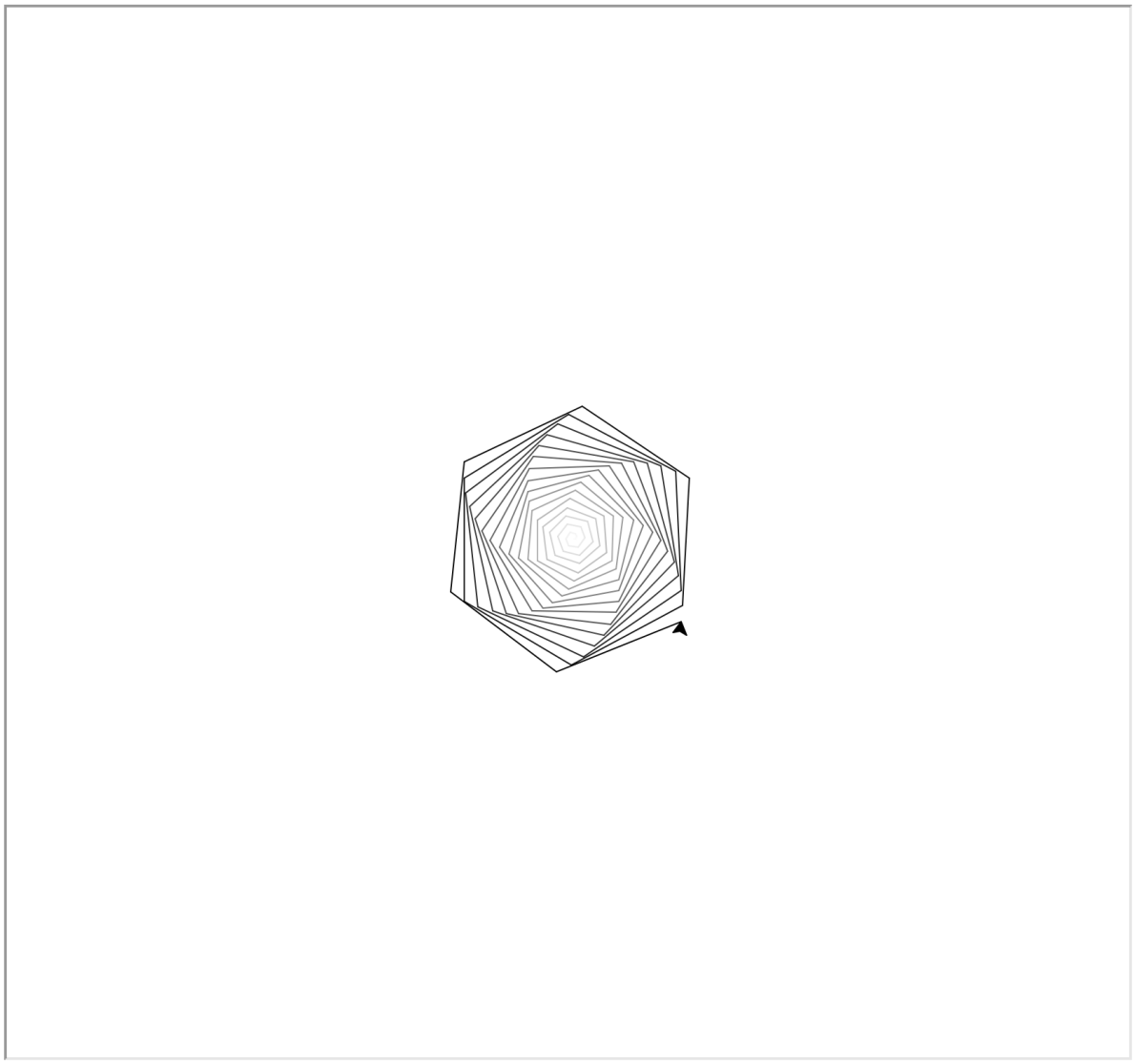
Deliverable: Homework03.py

References:

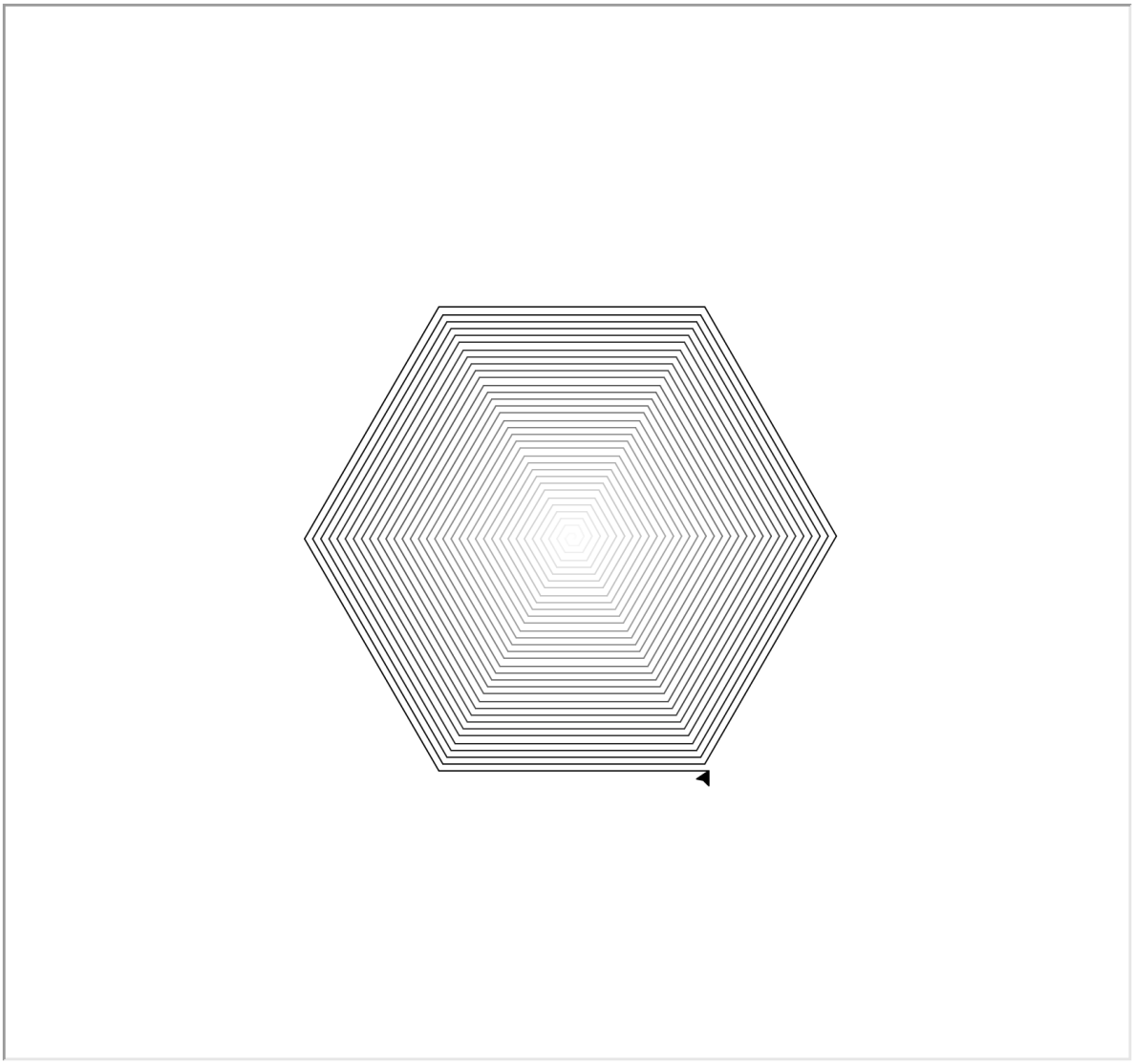
- https://runestone.academy/runestone/books/published/thinkcspy/PythonTurtle/toctree.html

Sample Input/Output:

```
$ python3 Homework03.py
Enter int angle: 59
Enter int range: 100
Click turtle screen to exit...
```



```
$ python3 Homework03.py
Enter int angle: 60
Enter int range: 200
Click turtle screen to exit...
```



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Homework 04

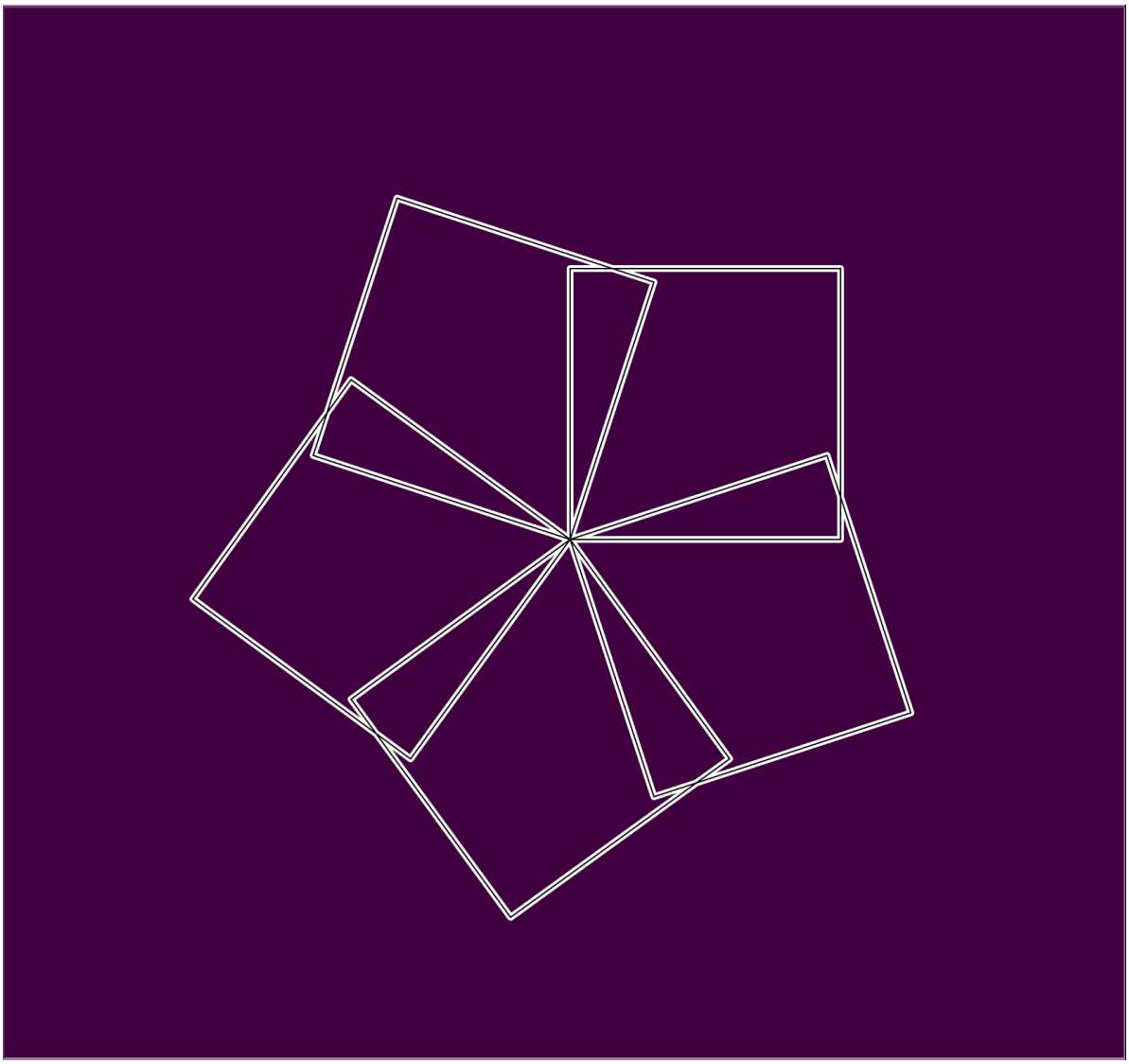
Deliverable: Homework04.py

References:

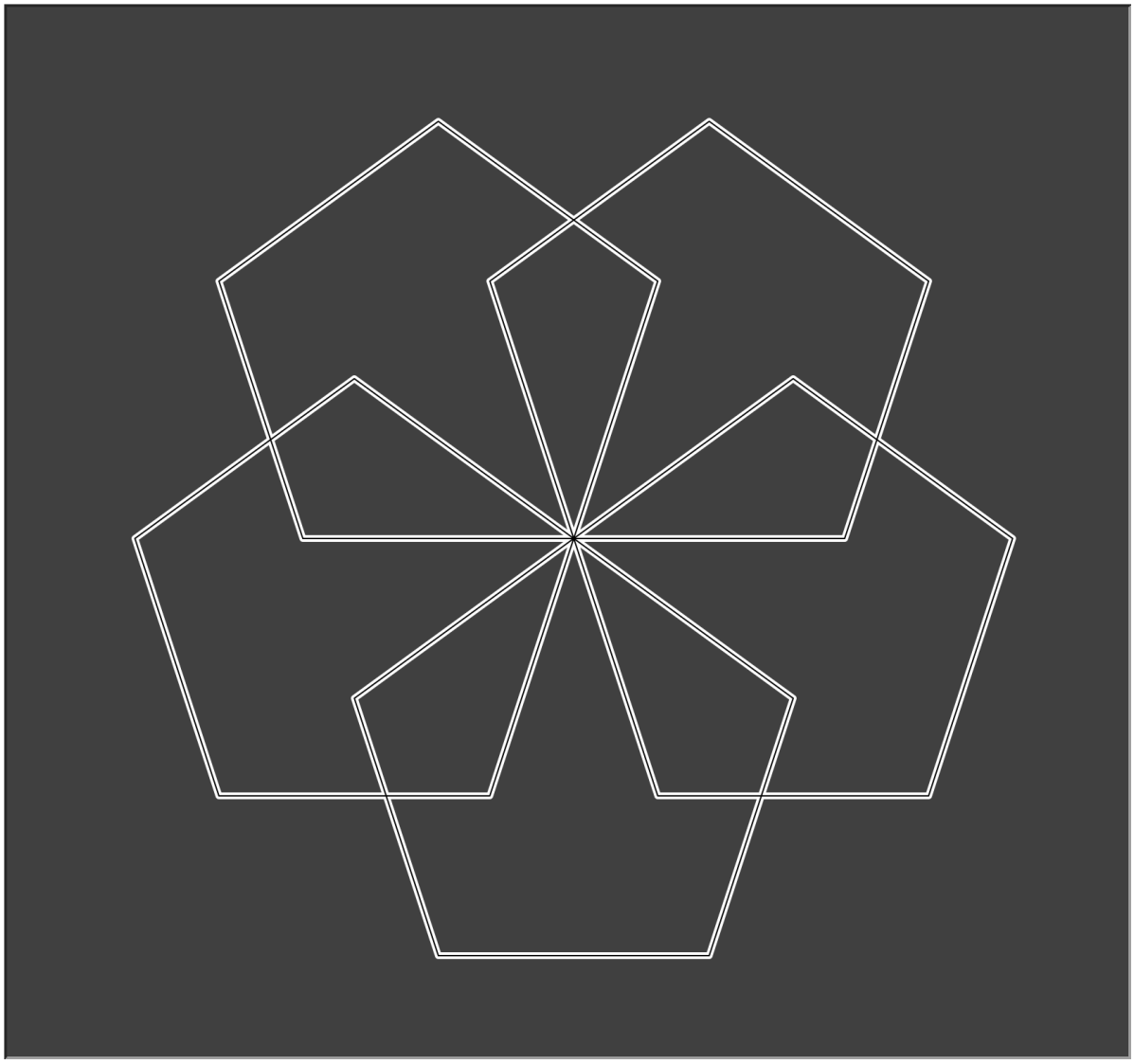
- https://runestone.academy/runestone/books/published/thinkcspy/PythonModules/toctree.html
- https://runestone.academy/runestone/books/published/thinkcspy/Functions/toctree.html

Sample Input/Output:

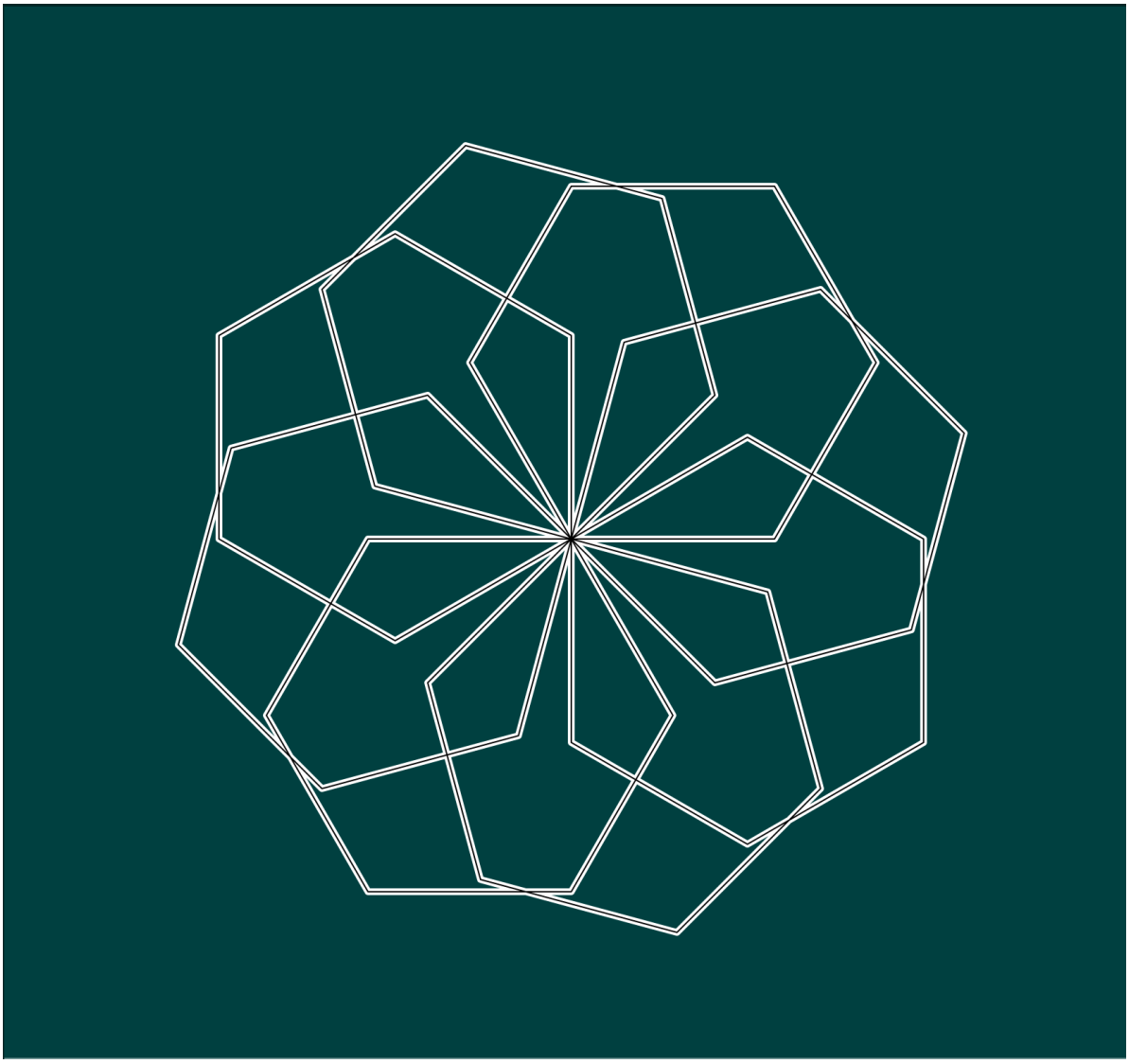
```
$ python3 Homework04.py
Random background color is: ( 0.25 , 0.0 , 0.25 )
Enter rotational copies: 5
Enter sides per polygon: 4
Enter edge pixel length: 200
Click turtle screen to exit...
```



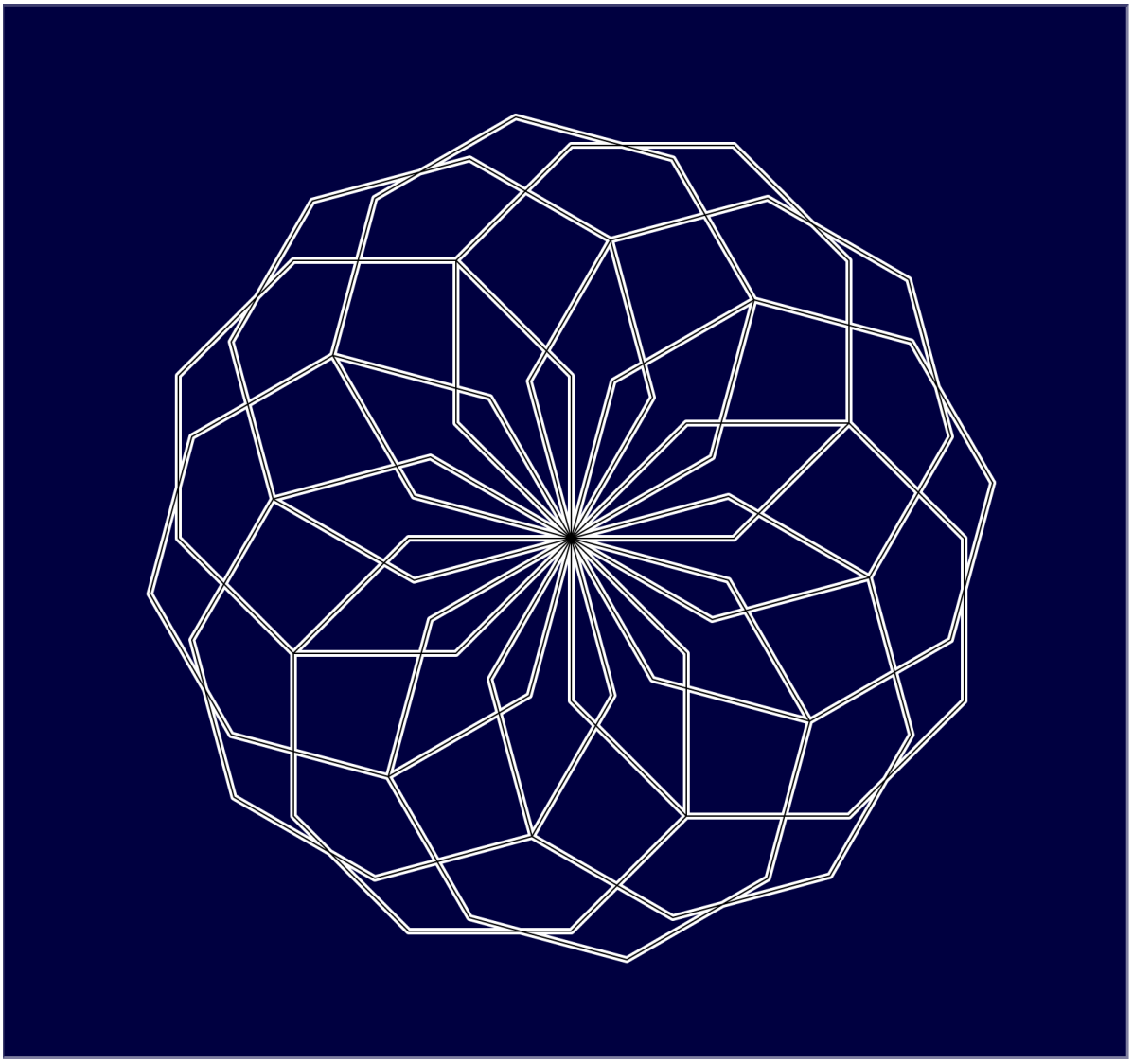
```
$ python3 Homework04.py
Random background color is: ( 0.25 , 0.25 , 0.25 )
Enter rotational copies: 5
Enter sides per polygon: 5
Enter edge pixel length: 200
Click turtle screen to exit...
```



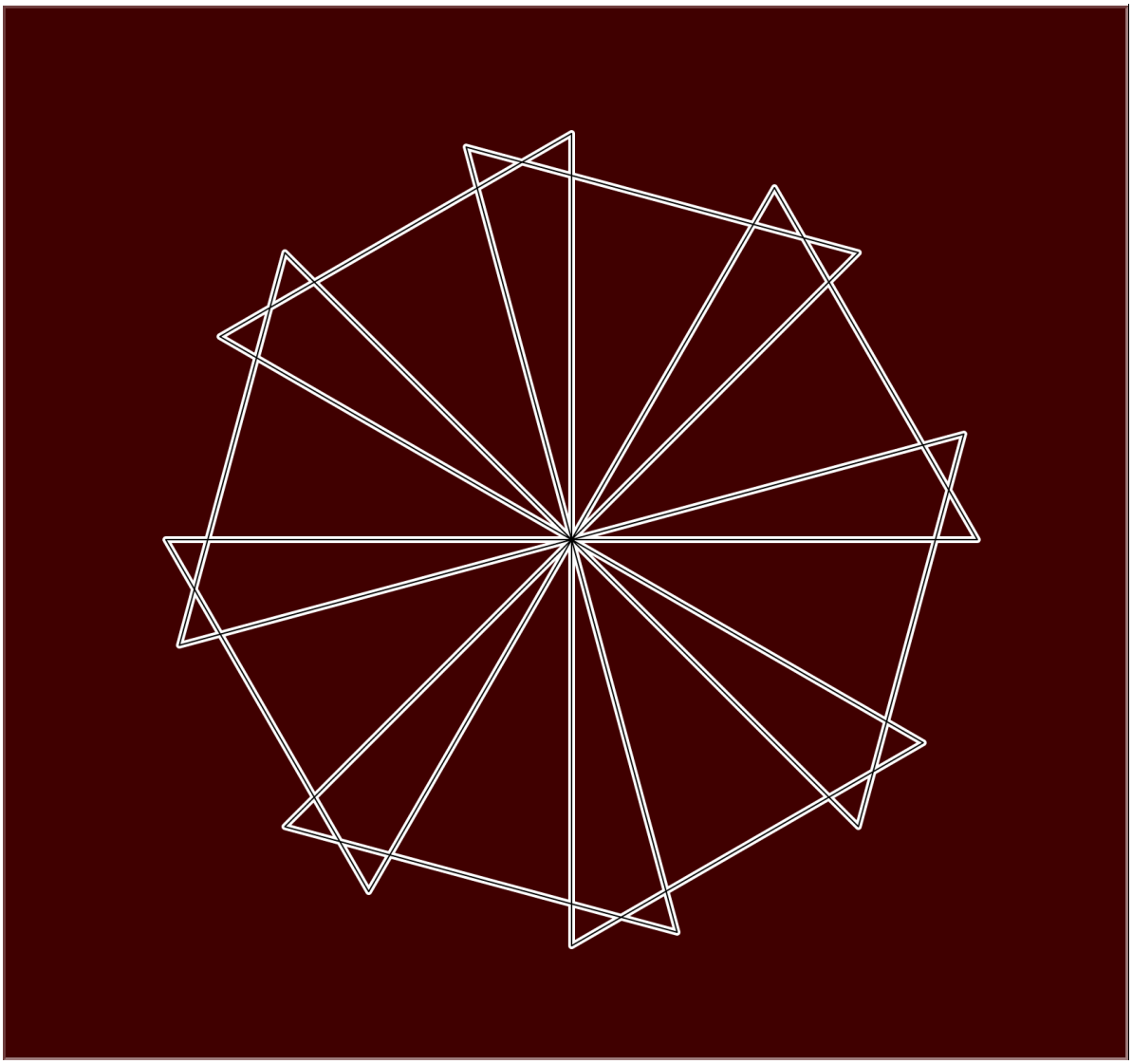
```
$ python3 Homework04.py
Random background color is: ( 0.0 , 0.25 , 0.25 )
Enter rotational copies: 8
Enter sides per polygon: 6
Enter edge pixel length: 150
Click turtle screen to exit...
```



```
$ python3 Homework04.py
Random background color is: ( 0.0 , 0.0 , 0.25 )
Enter rotational copies: 12
Enter sides per polygon: 8
Enter edge pixel length: 120
Click turtle screen to exit...
```



```
$ python3 Homework04.py
Random background color is: ( 0.25 , 0.0 , 0.0 )
Enter rotational copies: 8
Enter sides per polygon: 3
Enter edge pixel length: 300
Click turtle screen to exit...
```



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Homework 05

Deliverable: Homework05.py

References:

- https://runestone.academy/runestone/books/published/thinkcspy/Selection/toctree.html
- https://runestone.academy/runestone/books/published/thinkcspy/MoreAboutIteration/toctree.html

Sample Input/Output:

```
$ python3 Homework05.py
Enter positive integer radius (0 to quit): 1
  *
* * *
  *

Enter positive integer radius (0 to quit): 2
    *
  * * *
* * * * *
  * * *
    *

Enter positive integer radius (0 to quit): 3
      *
  * * * * *
  * * * * *
* * * * * * *
  * * * * *
  * * * * *
      *

Enter positive integer radius (0 to quit): 4
        *
    * * * * *
  * * * * * * *
  * * * * * * *
* * * * * * * * *
  * * * * * * *
  * * * * * * *
    * * * * *
        *
```

```
Enter positive integer radius (0 to quit): 5
          *
    * * * * * * *
  * * * * * * * * *
  * * * * * * * * *
  * * * * * * * * *
* * * * * * * * * * *
  * * * * * * * * *
  * * * * * * * * *
  * * * * * * * * *
    * * * * * * *
          *

Enter positive integer radius (0 to quit): 10
                    *
            * * * * * * * * *
        * * * * * * * * * * * * *
      * * * * * * * * * * * * * * *
    * * * * * * * * * * * * * * * * *
    * * * * * * * * * * * * * * * * *
  * * * * * * * * * * * * * * * * * * *
  * * * * * * * * * * * * * * * * * * *
  * * * * * * * * * * * * * * * * * * *
  * * * * * * * * * * * * * * * * * * *
* * * * * * * * * * * * * * * * * * * * *
  * * * * * * * * * * * * * * * * * * *
  * * * * * * * * * * * * * * * * * * *
  * * * * * * * * * * * * * * * * * * *
  * * * * * * * * * * * * * * * * * * *
    * * * * * * * * * * * * * * * * *
    * * * * * * * * * * * * * * * * *
      * * * * * * * * * * * * * * *
        * * * * * * * * * * * * *
            * * * * * * * * *
                    *

Enter positive integer radius (0 to quit): 0

Quitting program now...
```

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Homework 06

Deliverable: Homework06.py

References:

- https://runestone.academy/runestone/books/published/thinkcspy/Strings/toctree.html

Sample Input/Output:

```
$ python3 Homework06.py
Enter (^madlib^): Hey diddle diddle, the (^noun^) and the fiddle, The
(^noun^) (^past-tense verb^) over the moon, The (^adjective^) (^noun^)
(^past-tense verb^) to see such fun, And the (^noun^) (^past-tense verb^)
away with the spoon!
Enter number of blank lines to hide the madlib: 5




Enter noun: laptop
Enter noun: student
Enter past-tense verb: slept
Enter adjective: Python-infested
Enter noun: professor
Enter past-tense verb: programmed
Enter noun: homework
Enter past-tense verb: exploded
Hey diddle diddle, the laptop and the fiddle, The student slept over the
moon, The Python-infested professor programmed to see such fun, And the
homework exploded away with the spoon!


$ python3 Homework06.py
Enter (^madlib^): (^Proper Name^) is (^adjective^) whenever (^phrase^). This
^)example(^ has typos.
Enter number of blank lines to hide the madlib: 1


Enter Proper Name: Dr. E.
Enter adjective: shiny
Enter phrase: his head catches the light
Dr. E. is shiny whenever his head catches the light. This ^)example(^ has
typos.
```

CS121P Homework Assignments

Homework 07

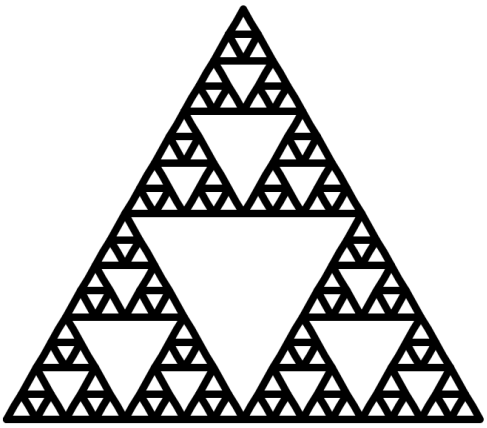
Deliverable: Homework07.py

References:

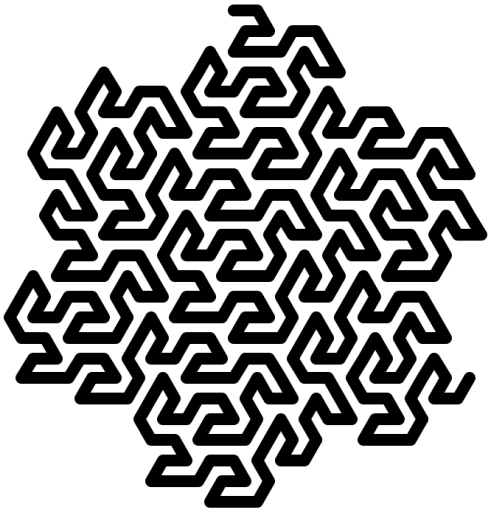
- https://runestone.academy/runestone/books/published/thinkcspy/Strings/toctree.html
- https://runestone.academy/runestone/books/published/thinkcspy/Lists/toctree.html

Sample Input/Output:

```
$ python3 Homework07.py

Symbols: 'F'=MoveForward,
         'B'=MoveBackward,
         '+'=TurnRight,
         '-'=TurnLeft,
         '>'=PenWider,
         '<'=PenNarrower,
         '['=StateSave,
         ']'=StateRestore,
         'ACDEGHIJKLMNOPQRSTUVWXYZ'=TurtleIgnores

Enter iterations: 4
Enter length: 10
Enter angle: 60
Enter axiom: >>>>FXF--FF--FF
Enter new rule 'LHS -> RHS' (empty line to exit): X -> --FXF++FXF++FXF--
Enter new rule 'LHS -> RHS' (empty line to exit): F -> FF
Enter new rule 'LHS -> RHS' (empty line to exit):
>>>>FFFFFFFFFFFFFFF--FFFFFFFF--FFFF--FF--FXF++FXF++FXF--FF++FF--
FXF++FXF++FXF--FF++FF--FXF++FXF++FXF--FF--FFFF++FFFF--FF--FXF++FXF++FXF--
FF++FF--FXF++FXF++FXF--FF++FF--FXF++FXF++FXF--FF--FFFF++FFFF--FF--
FXF++FXF++FXF--FF++FF--FXF++FXF++FXF--FF++FF--FXF++FXF++FXF--FF--FFFF--
FFFFFFFF++FFFFFFFF--FFFF--FF--FXF++FXF++FXF--FF++FF--FXF++FXF++FXF--FF++FF--
FXF++FXF++FXF--FF--FFFF++FFFF--FF--FXF++FXF++FXF--FF++FF--FXF++FXF++FXF--
FF++FF--FXF++FXF++FXF--FF--FFFF++FFFF--FF--FXF++FXF++FXF--FF++FF--
FXF++FXF++FXF--FF++FF--FXF++FXF++FXF--FF--FFFF--FFFFFFFF++FFFFFFFF--FFFF--FF-
-FXF++FXF++FXF--FF++FF--FXF++FXF++FXF--FF++FF--FXF++FXF++FXF--FF--FFFF++FFFF-
-FF--FXF++FXF++FXF--FF++FF--FXF++FXF++FXF--FF++FF--FXF++FXF++FXF--FF--
FFFF++FFFF--FF--FXF++FXF++FXF--FF++FF--FXF++FXF++FXF--FF++FF--FXF++FXF++FXF--
FF--FFFF--FFFFFFFF--FFFFFFFFFFFFFFFF--FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF--
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
```



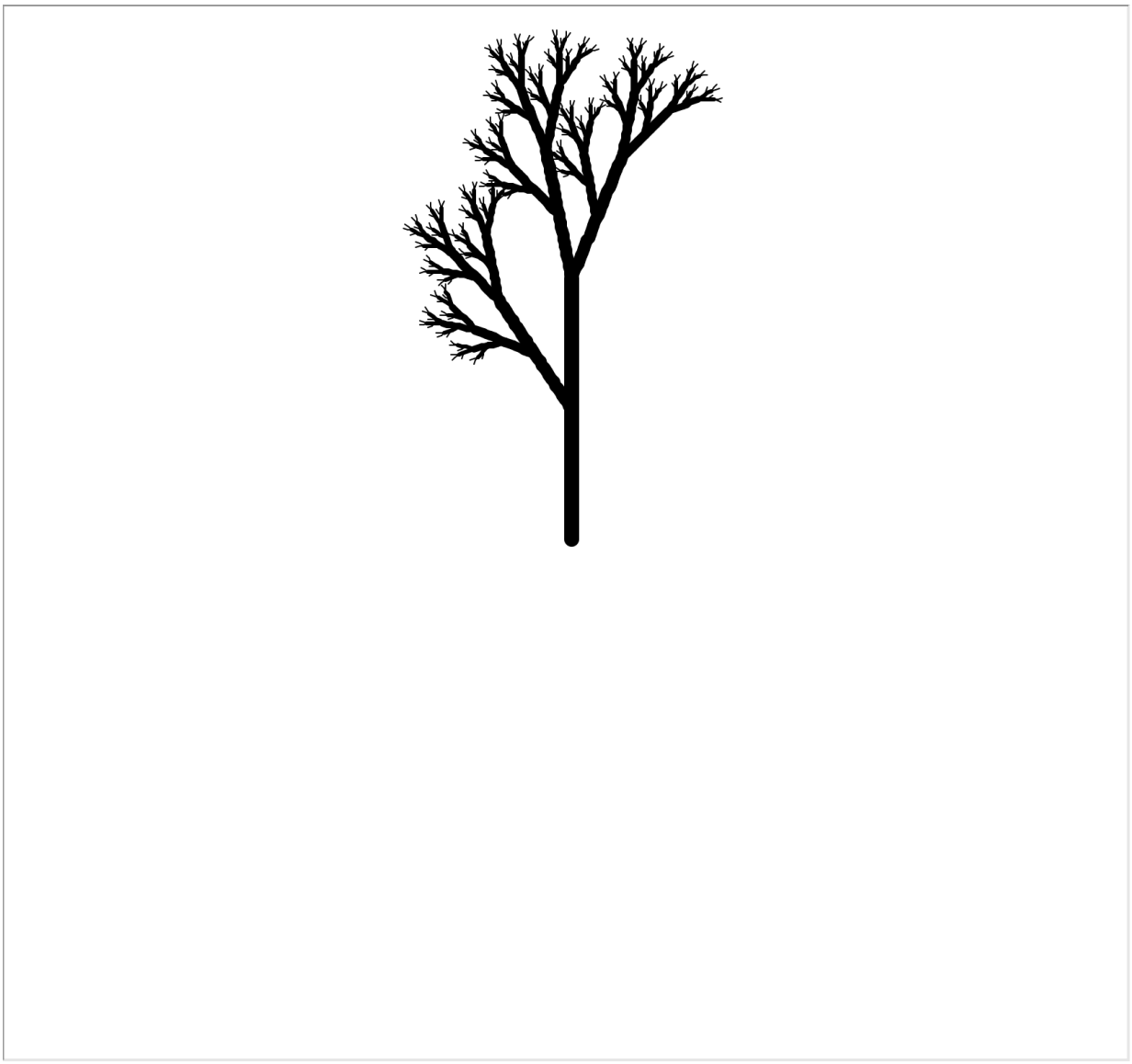
```
$ python3 Homework07.py


Symbols: 'F'=MoveForward,
         'B'=MoveBackward,
         '+'=TurnRight,
         '-'=TurnLeft,
         '>'=PenWider,
         '<'=PenNarrower,
         '['=StateSave,
         ']'=StateRestore,
         'ACDEGHIJKLMNOPQRSTUVWXYZ'=TurtleIgnores

Enter iterations: 3
Enter length: 16
Enter angle: 60
Enter axiom: >>>>>>>FX
Enter new rule 'LHS -> RHS' (empty line to exit): X -> X+YF++YF-FX--FXFX-YF+
Enter new rule 'LHS -> RHS' (empty line to exit): Y -> -FX+YFYF++YF+FX--FX-Y
Enter new rule 'LHS -> RHS' (empty line to exit):
>>>>>>>FX+YF++YF-FX--FXFX-YF++-FX+YFYF++YF+FX--FX-YF++-FX+YFYF++YF+FX--FX-YF-
FX+YF++YF-FX--FXFX-YF+--FX+YF++YF-FX--FXFX-YF+FX+YF++YF-FX--FXFX-YF+--
FX+YFYF++YF+FX--FX-YF++-FX+YF++YF-FX--FXFX-YF++-FX+YFYF++YF+FX--FX-YF-
FX+YFYF++YF+FX--FX-YF++-FX+YFYF++YF+FX--FX-YF+FX+YF++YF-FX--FXFX-YF+--
FX+YF++YF-FX--FXFX-YF+--FX+YFYF++YF+FX--FX-YF++-FX+YF++YF-FX--FXFX-YF++-
FX+YFYF++YF+FX--FX-YF-FX+YFYF++YF+FX--FX-YF++-FX+YFYF++YF+FX--FX-
YF+FX+YF++YF-FX--FXFX-YF+--FX+YF++YF-FX--FXFX-YF+--FX+YFYF++YF+FX--FX-YF-
FX+YF++YF-FX--FXFX-YF++-FX+YFYF++YF+FX--FX-YF++-FX+YFYF++YF+FX--FX-YF-
FX+YF++YF-FX--FXFX-YF+--FX+YF++YF-FX--FXFX-YF+FX+YF++YF-FX--FXFX-YF+--
FX+YFYF++YF+FX--FX-YF+--FX+YF++YF-FX--FXFX-YF++-FX+YFYF++YF+FX--FX-YF++-
FX+YFYF++YF+FX--FX-YF-FX+YF++YF-FX--FXFX-YF+--FX+YF++YF-FX--FXFX-
YF+FX+YF++YF-FX--FXFX-YF+--FX+YFYF++YF+FX--FX-YF+FX+YF++YF-FX--FXFX-YF++-
FX+YFYF++YF+FX--FX-YF++-FX+YFYF++YF+FX--FX-YF-FX+YF++YF-FX--FXFX-YF+--
FX+YF++YF-FX--FXFX-YF+FX+YF++YF-FX--FXFX-YF+--FX+YFYF++YF+FX--FX-YF+--
FX+YF++YF-FX--FXFX-YF++-FX+YFYF++YF+FX--FX-YF-FX+YFYF++YF+FX--FX-YF++-
FX+YFYF++YF+FX--FX-YF+FX+YF++YF-FX--FXFX-YF+--FX+YF++YF-FX--FXFX-YF+--
FX+YFYF++YF+FX--FX-YF+
```



```
$ python3 Homework07.py

Symbols: 'F'=MoveForward,
         'B'=MoveBackward,
         '+'=TurnRight,
         '-'=TurnLeft,
         '>'=PenWider,
         '<'=PenNarrower,
         '['=StateSave,
         ']'=StateRestore,
         'ACDEGHIJKLMNOPQRSTUVWXYZ'=TurtleIgnores

Enter iterations: 6
Enter length: 3
Enter angle: 11.25
Enter axiom: --------X
Enter new rule 'LHS -> RHS' (empty line to exit): X -> F[---X]F[++X]-X
Enter new rule 'LHS -> RHS' (empty line to exit): F -> >>FF<<
Enter new rule 'LHS -> RHS' (empty line to exit):
-------
->>>>>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<
<<>>>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<<
<<<[--
->>>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<<<
[--->>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<[--->>>>FF<<>>FF<<<<[--->>FF<<[---F[-
--X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[--
-X]F[++X]-X]->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---
X]F[++X]-X]>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<[++>>>>FF<<>>FF<<<<[--
->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[--
-X]F[++X]-X]->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---
X]F[++X]-X]->>>>FF<<>>FF<<<<[--->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---
X]F[++X]-X]-F[---X]F[++X]-X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-
X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-X]->>FF<<[---F[---X]F[++X]-
X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<<
<[++>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<[--->>>>FF<<>>FF<<<<[--->>FF<<[---F[-
--X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[--
-X]F[++X]-X]->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---
X]F[++X]-X]>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<[++>>>>FF<<>>FF<<<<[--
->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[--
-X]F[++X]-X]->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---
X]F[++X]-X]->>>>FF<<>>FF<<<<[--->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---
X]F[++X]-X]-F[---X]F[++X]-X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-
X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-X]->>FF<<[---F[---X]F[++X]-
X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]->>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<[--->>>>FF<<>>FF<<<<[--->>FF<<[---F[--
-X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[--
-X]F[++X]-X]->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---
X]F[++X]-X]>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<[++>>>>FF<<>>FF<<<<[--
->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[--
-X]F[++X]-X]->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---
```

```
X]F[++X]-X]->>>>FF<<>>FF<<<<[--->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---
X]F[++X]-X]-F[---X]F[++X]-X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-
X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-X]->>FF<<[---F[---X]F[++X]-
X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<
<<<>>>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<
<<<<[++>>>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<
<<<<<<[--->>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<[--->>>>FF<<>>FF<<<<[--
->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[--
-X]F[++X]-X]->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---
X]F[++X]-X]>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<[++>>>>FF<<>>FF<<<<[--
->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[--
-X]F[++X]-X]->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---
X]F[++X]-X]->>>>FF<<>>FF<<<<[--->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---
X]F[++X]-X]-F[---X]F[++X]-X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-
X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-X]->>FF<<[---F[---X]F[++X]-
X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<<
<[++>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<[--->>>>FF<<>>FF<<<<[--->>FF<<[---F[-
--X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[--
-X]F[++X]-X]->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---
X]F[++X]-X]>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<[++>>>>FF<<>>FF<<<<[--
->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[--
-X]F[++X]-X]->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---
X]F[++X]-X]->>>>FF<<>>FF<<<<[--->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---
X]F[++X]-X]-F[---X]F[++X]-X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-
X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-X]->>FF<<[---F[---X]F[++X]-
X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]->>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<[--->>>>FF<<>>FF<<<<[--->>FF<<[---F[--
-X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[--
-X]F[++X]-X]->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---
X]F[++X]-X]>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<[++>>>>FF<<>>FF<<<<[--
->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[--
-X]F[++X]-X]->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---
X]F[++X]-X]->>>>FF<<>>FF<<<<[--->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---
X]F[++X]-X]-F[---X]F[++X]-X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-
X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-X]->>FF<<[---F[---X]F[++X]-
X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]->>>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<
<<[--->>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<[--->>>>FF<<>>FF<<<<[--->>FF<<[---
F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[--
-X]F[++X]-X]->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---
X]F[++X]-X]>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<[++>>>>FF<<>>FF<<<<[--
->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[--
-X]F[++X]-X]->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---
X]F[++X]-X]->>>>FF<<>>FF<<<<[--->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---
X]F[++X]-X]-F[---X]F[++X]-X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-
X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-X]->>FF<<[---F[---X]F[++X]-
X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
```

```
X]>>>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<<
<[++>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<[--->>>>FF<<>>FF<<<<[--->>FF<<[---F[-
--X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[--
-X]F[++X]-X]->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---
X]F[++X]-X]>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<[++>>>>FF<<>>FF<<<<[--
->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[--
-X]F[++X]-X]->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---
X]F[++X]-X]->>>>FF<<>>FF<<<<[--->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---
X]F[++X]-X]-F[---X]F[++X]-X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-
X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-X]->>FF<<[---F[---X]F[++X]-
X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]->>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<[--->>>>FF<<>>FF<<<<[--->>FF<<[---F[--
-X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[--
-X]F[++X]-X]->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---
X]F[++X]-X]>>>>>>FF<<>>FF<<<<>>>>FF<<>>FF<<<<<<[++>>>>FF<<>>FF<<<<[--
->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-
X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[--
-X]F[++X]-X]->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---X]F[++X]-X]-F[---
X]F[++X]-X]->>>>FF<<>>FF<<<<[--->>FF<<[---F[---X]F[++X]-X]>>FF<<[++F[---
X]F[++X]-X]-F[---X]F[++X]-X]>>>>FF<<>>FF<<<<[++>>FF<<[---F[---X]F[++X]-
X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-X]->>FF<<[---F[---X]F[++X]-
X]>>FF<<[++F[---X]F[++X]-X]-F[---X]F[++X]-X
```



Homework 08

Deliverable: Homework08.py

References:

- https://runestone.academy/runestone/books/published/thinkcspy/Files/toctree.html
- https://runestone.academy/runestone/books/published/thinkcspy/Dictionaries/toctree.html
- https://runestone.academy/runestone/books/published/thinkcspy/Exceptions/toctree.html

Sample Input/Output:

```
$ python3 Homework08.py
Reading input file "students.txt"...
Error: students.txt does not exist or it can't be opened for input.
Program exiting now...

$ python3 Homework08.py
Reading input file "students.txt"...
Input processed.
Writing output file "classes.txt"...
Error: classes.txt can't be opened for output.
Program exiting now...

$ python3 Homework08.py
Reading input file "students.txt"...
Input processed.
Writing output file "classes.txt"...
Output processed.
Program exiting now...
```

Sample Input File:

- students.txt

```
Bunyan, John: CS121P MA253 CS361A
Calvin, John: CS132S CS202H CS312N CS342D CS362N
Cranmer, Thomas: CS301A CS321O CS321P CS361M
Edwards, Jonathan: CS322E CS490I CS492S CS362V
Henry, Matthew: CS121P MA121 MA253
Knox, John: CS132S CS202H CS312N CS342D CS322X
Latimer, Hugh: CS301A CS321O CS321P CS341W CS351S
Lloyd-Jones, Martyn: CS322E CS490I CS492S
Luther, Martin: CS121P MA253 ACC210 MGT310
Owen, John: CS132S CS202H CS312N CS342D
Ridley, Nicolas: CS301A CS321O CS321P CS311C
Rodgers, John: CS322E CS490I CS492S CS392M
Rutherford, Samuel: MA121 CS341W CS351S CS361A
Spurgeon, Charles: ACC210 MGT310 CS311C CS361M
Tyndale, William: MA122 MA262 CS322X CS362N
Whitefield, George: MA262 CS362V CS392M
```

Sample Output File:

- classes.txt

```
ACC210:
Luther, Martin
Spurgeon, Charles

CS121P:
Bunyan, John
Henry, Matthew
Luther, Martin

CS132S:
Calvin, John
Knox, John
Owen, John

CS202H:
Calvin, John
Knox, John
Owen, John

CS301A:
Cranmer, Thomas
Latimer, Hugh
Ridley, Nicolas

CS311C:
Ridley, Nicolas
Spurgeon, Charles

CS312N:
Calvin, John
Knox, John
Owen, John

CS321O:
Cranmer, Thomas
Latimer, Hugh
Ridley, Nicolas

CS321P:
Cranmer, Thomas
Latimer, Hugh
Ridley, Nicolas

CS322E:
Edwards, Jonathan
Lloyd-Jones, Martyn
Rodgers, John

CS322X:
Knox, John
Tyndale, William

CS341W:
Latimer, Hugh
Rutherford, Samuel
```

CS121P Homework Assignments

CS342D:
Calvin, John
Knox, John
Owen, John

CS351S:
Latimer, Hugh
Rutherford, Samuel

CS361A:
Bunyan, John
Rutherford, Samuel

CS361M:
Cranmer, Thomas
Spurgeon, Charles

CS362N:
Calvin, John
Tyndale, William

CS362V:
Edwards, Jonathan
Whitefield, George

CS392M:
Rodgers, John
Whitefield, George

CS490I:
Edwards, Jonathan
Lloyd-Jones, Martyn
Rodgers, John

CS492S:
Edwards, Jonathan
Lloyd-Jones, Martyn
Rodgers, John

MA121:
Henry, Matthew
Rutherford, Samuel

MA122:
Tyndale, William

MA253:
Bunyan, John
Henry, Matthew
Luther, Martin

MA262:
Tyndale, William
Whitefield, George

MGT310:
Luther, Martin
Spurgeon, Charles

Homework 09

Deliverable: Homework09.py

References:

- https://runestone.academy/runestone/books/published/pythonds/Introduction/ObjectOrientedProgramminginPythonDefiningClasses.html

Sample Input/Output:

```
$ python3 Homework09.py
******************************
Printing main() source code:
******************************
def main():
    print("*" * 30 + "\nPrinting main() source code:\n" + "*" * 30)
    import inspect
    print(str(inspect.getsource(main)))
    print("*" * 30 + "\nPrinting main() source output:\n" + "*" * 30)
    f1 = Fraction(1, 2)
    f2 = Fraction(-7, 3)
    f3 = Fraction(-1, 4)
    f4 = Fraction(21, 5)
    f5 = f1 + f2 + f3 + f4

    print("f1: " + str(f1))
    print("f2: " + str(f2))
    print("f3: " + str(f3))
    print("f4: " + str(f4))
    print("f5: " + str(f5))
    print("isinstance(f5, Fraction): " + str(isinstance(f5, Fraction)))
    print("isinstance(f5, MixedFraction): " + str(isinstance(f5, MixedFraction)))

    m1 = MixedFraction(f1)
    m2 = MixedFraction(f2)
    m3 = MixedFraction(f3)
    m4 = MixedFraction(f4)
    m5 = m1 + m2 + m3 + m4

    print("m1: " + str(m1))
    print("m2: " + str(m2))
    print("m3: " + str(m3))
    print("m4: " + str(m4))
    print("m5: " + str(m5))
    print("isinstance(m5, Fraction): " + str(isinstance(m5, Fraction)))
    print("isinstance(m5, MixedFraction): " + str(isinstance(m5, MixedFraction)))

    print("(f5==m5) : " + str(f5==m5))

******************************
Printing main() source output:
******************************
f1: 1/2
f2: -7/3
f3: -1/4
f4: 21/5
f5: 127/60
isinstance(f5, Fraction): True
isinstance(f5, MixedFraction): False
m1: 0_(1/2)
m2: -2_(1/3)
m3: -0_(1/4)
m4: 4_(1/5)
m5: 2_(7/60)
isinstance(m5, Fraction): True
isinstance(m5, MixedFraction): True
(f5==m5) : True
```